Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

VISAI

16th INTERNATIONAL PROJECT COMPETITION & EXHIBITION
24th FEB 2026

Baraninathan S P **Deepak Sudalai M** **Ebin Jebaraj L**

HACKATHON/Name of the Company: Renault Nissan Technology

Problem Statement No: 3

College Name : Francis Xavier Engineering College

Reference Number : PST260052

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Problem Statement

Urban flooding is a frequent and severe problem in cities like Chennai during monsoon seasons. Existing flood monitoring systems are costly and limited in coverage, resulting in delayed warnings and inefficient emergency response.

Although cities already have extensive CCTV and traffic camera networks, these are not effectively used for real-time flood detection.

There is a need for a **low-cost, scalable, real-time flood detection system** that can utilize existing CCTV infrastructure to provide early alerts and support disaster management authorities.

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Concept of the solution

The proposed solution uses **AI-based computer vision** to analyze live feeds from existing CCTV and traffic cameras to detect flood conditions.

The system identifies **rising water levels, submerged roads, and stalled vehicles** in real time.

Detected flood data is visualized on a **central dashboard** and alerts are sent to authorities and the public through **SMS and mobile notifications**.

This approach enables **early warning, faster response, and improved public safety** without deploying new hardware.

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Novelty / Scope of Solution

Novelty:
•Uses **already-installed CCTV cameras**, avoiding expensive sensors.
•Works effectively with **low-resolution and angled camera feeds**.
•AI model trained specifically for **Indian urban flood scenarios**.
•Real-time alerts integrated with **disaster response systems**.

Scope:
•Can be deployed city-wide and extended to other flood-prone cities.
•Useful for traffic diversion, emergency routing, and evacuation planning.
•Supports long-term flood pattern analysis for urban planning.

Benchmark:
Compared to traditional water-level sensors, this solution offers **wider coverage, lower cost, and faster deployment**.

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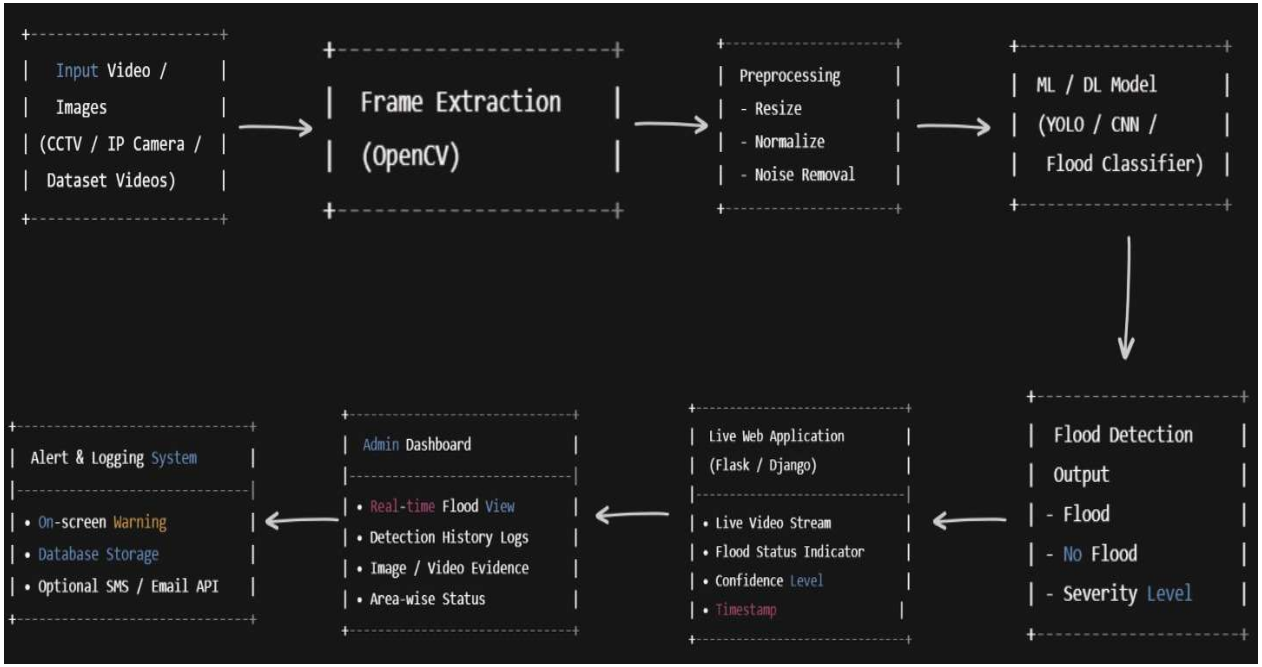
Technical Description

CCTV Output

Block Diagram



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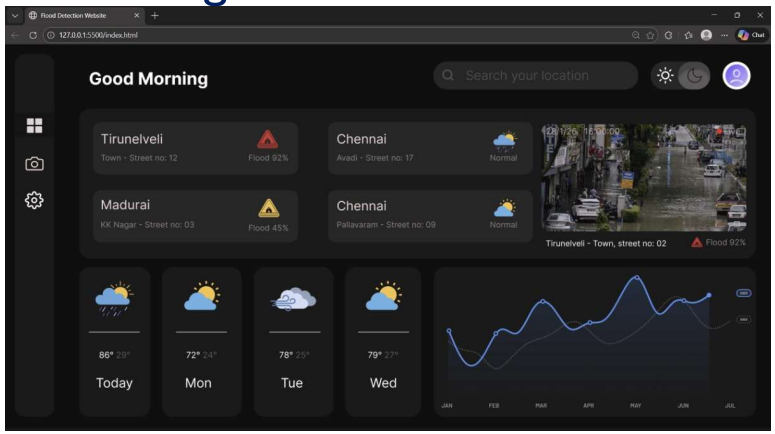
Implementation Plan/Working Model



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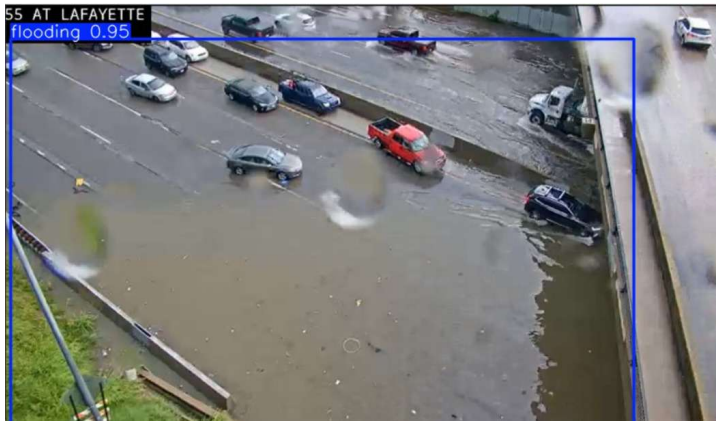
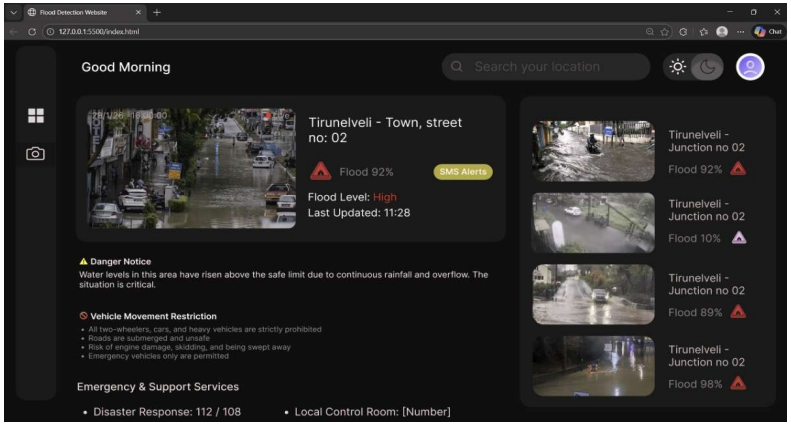
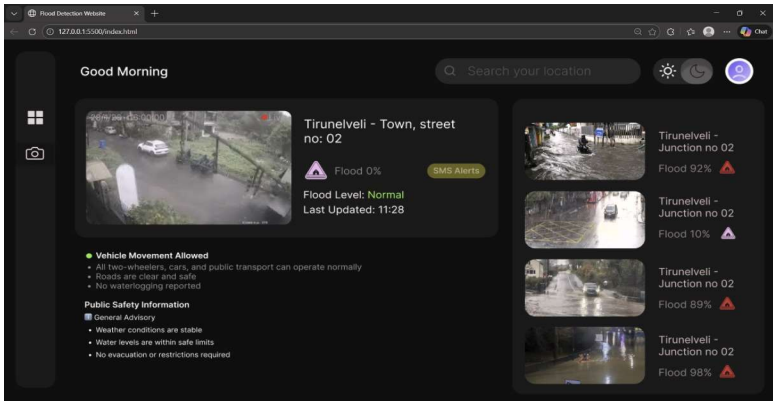
Validation / Testing / Analysis







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References

•YOLO Object Detection Framework - https://universe.roboflow.com/yldz-teknik-niversitesi-ld9fy/flood-detection-3susv/dataset/2

•OpenCV – Webcam & Video Processing - https://docs.opencv.org/4.x/dd/d43/tutorial_py_video_display.html

•Computer Vision & Deep Learning - https://cs231n.github.io/convolutional-networks/

•Smart City Surveillance

Pros and Cons of the solution

Pros

- Cost-effective and scalable
- No new hardware required
- Real-time monitoring website and alerts
- Wide city coverage
- Faster emergency response

Cons

- Performance affected by extreme visibility issues
- Requires stable network connectivity

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THANK YOU